

When the Shock Hits the Top: Artificial Intelligence, the Exchequer, and the Limits of UK Redistribution*

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July 2026 — preliminary draft

Abstract

We examine how the adoption of artificial intelligence could affect employment, household incomes, income inequality and the public finances in the United Kingdom, replicating the scenario framework that Doorley et al. (2026) developed for Ireland. Occupation-level AI exposure (the complementarity-adjusted AIOE) is attached to Family Resources Survey adults through their SOC2020 occupation; employment, wage and capital-return shocks are distributed across occupations by relative exposure; and the resulting counterfactual income distributions are evaluated with the PolicyEngine UK tax-benefit microsimulation model. In the central scenario — 7 per cent job displacement, a 2.6 per cent wage gain for those who remain, and a 0.4 percentage-point rise in the return to capital — the annual Exchequer cost is £18.6 billion ($\pm$ £1.7 billion across displacement draws), the before-housing-costs poverty rate rises by around 2 percentage points, and the Gini index of equivalised household disposable income rises by 1.2–1.3 percentage points. Displacement risk is concentrated in higher-income households, yet inequality rises in every one of 66 scenarios considered. We extend the framework along the age dimension, implementing recent seniority-biased evidence as explicit scenarios.

Executive summary

The United Kingdom enters the era of generative artificial intelligence (AI) with an unusual combination of features: a workforce among the most AI-exposed in the advanced economies, concentrated in the professional and managerial services that dominate its labour market, and a tax-benefit system that is among the most redistributive in Europe — but one designed to insure shocks that strike the *bottom* of the income distribution. This report asks what happens when those two facts collide. AI displacement risk, unlike every previous wave of automation, is concentrated at the top: in our central calibration, the share of workers at risk of displacement rises from 0.4 per cent in the bottom decile of household income to 5.0 per cent in the top. We trace this shock — together with wage gains for the workers AI complements and higher returns to capital — through the full UK tax and benefit system using open-source microsimulation

*This study applies the scenario architecture of Doorley, O'Connor, O'Shea and Tuda (2026), *Artificial intelligence and income inequality in Ireland*, ESRI/Department of Finance Jointly-published Report No. 16, to the United Kingdom, implemented in the open-source PolicyEngine UK microsimulation model. Code and aggregate results: <https://github.com/PolicyEngine/uk-ai-study>. Family Resources Survey microdata are accessed under the UK Data Service End User Licence and are not redistributed. Draft — not for citation.

(PolicyEngine UK on the Family Resources Survey 2024–25), across a grid of 66 scenarios spanning the range of current evidence. Rather than forecasting a single outcome, the results describe the shape of the risk.

A central and robust finding is that the risk of AI-driven job displacement is concentrated towards the top of the income distribution. Workers in occupations most exposed to AI—professional, managerial, and associate professional roles—are disproportionately found in higher-income households. In our central calibration, the share of workers at risk of transition rises steadily from 0.4 per cent of adults in the bottom decile of equivalised household disposable income to 5.0 per cent in the top decile. By contrast, the wage gains that accrue to workers who keep their jobs are modest and remarkably flat across the distribution, ranging only from about 2.5 to 2.7 per cent across deciles. Capital income gains, however, are heavily skewed: households in the top decile receive roughly 35 per cent of all interest and dividend income, so any AI-driven rise in returns to capital flows overwhelmingly to the top.

In our central scenario—7 per cent of exposed workers displaced, a 2.6 per cent wage gain for those who remain in work, and a 0.4 percentage point increase in the return on capital—the net fiscal cost to the Exchequer is £18.6 billion per year, as lost income tax and National Insurance receipts and higher benefit spending outweigh additional revenue from wage and capital gains. Poverty measured before housing costs rises by 2.1 percentage points and the Gini coefficient of disposable income increases by 1.3 percentage points (mean across displacement draws: +1.15). The most striking result of the full scenario analysis is its uniformity of direction: the Gini coefficient rises in every one of the 66 scenarios we examine, with increases ranging from 0.05 to 1.9 percentage points. Although displacement initially strikes better-off households, the combination of income losses spread through the distribution and capital gains concentrated at the very top means that AI raises measured inequality under every combination of assumptions we consider.

A structural blind spot of exposure-based frameworks — ours included — is age. Occupational exposure indices cannot distinguish a junior analyst from a senior one doing related work, yet the emerging evidence is that generative AI bears most heavily on the young: junior headcount contracts sharply at AI-adopting firms while senior staffing is untouched (Hosseini and Lichtinger, 2026; Klein Teeselink, 2025; Brynjolfsson et al., 2025). Because no exposure index can generate this seniority bias endogenously, we impose it: an age dimension added to the displacement mechanics, with the seniority-biased estimates from this literature implemented as explicit calibration scenarios. The exercise shows that frameworks without such a correction systematically understate the risk to young workers — in our exposure-only central scenario, prime-age workers face the highest displacement rates simply because they dominate exposed occupations.

Two policy implications follow. First, because displacement risk sits in occupations and households not traditionally served by active labour market policy, retraining and lifelong learning provision needs to reach mid-career professional workers, and to do so before rather than after displacement occurs. Second, the fiscal results highlight a structural vulnerability: the UK’s revenue base leans heavily on taxing labour income, so a shift in national income from labour to capital erodes receipts even if aggregate income holds up. Ensuring that the tax system captures a fair share of AI-driven capital returns—and that benefit rules cushion transitions without discouraging re-employment—will be central to making the gains from AI compatible with the UK’s redistributive settlement.

1 Introduction

The United Kingdom pairs one of the most AI-exposed workforces in the advanced economies with one of Europe’s most redistributive tax-benefit systems—and the two were not designed for each other. The UK’s services-intensive economy concentrates employment in the professional, financial, administrative and managerial occupations that score highest on standard measures of AI exposure (Felten et al., 2021; Eloundou et al., 2024; Pizzinelli et al., 2023), while its safety net—means-tested, tapered, and financed overwhelmingly by taxes on labour income—was built to insure shocks that strike the *bottom* of the income distribution. This report asks what happens when a labour-market shock arrives at the wrong end of the distribution the system was built for.

The question is new because the shock is new. Every previous wave of labour-saving technology in the UK hollowed out mid-skill, routine work (Autor et al., 2003), and the tax-benefit system proved effective at absorbing the household-income consequences (Doorley et al., 2023). Generative AI inverts the gradient: exposure rises with education and earnings, so displacement risk is concentrated in high-income households, while any capitalisation of AI’s gains flows to asset holders at the very top. A shock with this shape implicates the welfare state twice over—its insurance function faces claimants it rarely serves, and its revenue base sits precisely in the labour income the shock destroys.

Tracing the net effect requires microsimulation. AI is expected to move household incomes through three channels—displacement of workers in exposed occupations, wage gains for the workers it complements (Briggs and Kodnani, 2023), and higher returns to capital (Cazzaniga et al., 2024)—and workers at risk, workers who gain, and owners of capital occupy different positions in the income distribution. Only a full tax-benefit model reveals how taxes and transfers reshape the combined shock into disposable income, poverty and inequality outcomes, and what the residual claim on the exchequer amounts to.

Causal evidence on AI’s labour market effects remains scarce and recent, and no consensus estimate exists for the size of displacement, wage or capital effects over the coming years. We therefore follow a scenario-based approach rather than a forecasting one. We define plausible ranges for each of the three channels, drawing on the emerging empirical literature, and simulate a grid of scenarios covering their combinations. The output is not a prediction of what AI will do to UK incomes but a mapping of what it could do under transparent assumptions—and, importantly, an identification of results that hold across the entire scenario space.

Answering the question requires a framework with three properties: occupation-level exposure that can allocate shocks across the earnings distribution, scenario mechanics disciplined enough to compare dozens of parameter combinations, and a tax-benefit engine detailed enough to capture the UK’s means tests, tapers and marginal rates. The scenario architecture recently developed for Ireland by Doorley et al. (2026) provides the first two, and we adopt it as the best available design for this class of question; for the third we use PolicyEngine UK (v2.89.2), an open-source microsimulation model of the UK tax and benefit system, running on the Family Resources Survey 2024–25 with AI exposure assigned to SOC2020 occupations (Felten et al., 2021; Tolan et al., 2021; Eloundou et al., 2024). The open-source implementation is itself a contribution: every result in this report can be regenerated from public code, where comparable exercises rest on closed models and restricted microdata. Exposed workers face a probability of displacement, surviving workers receive a complementarity-graded wage premium, returns to household capital income rise, and the tax-benefit model translates these market-income shocks into disposable income, poverty and inequality outcomes.

One structural limitation of every exposure-based framework—including the one we adopt—motivates our main methodological extension. Occupational exposure indices are age-blind: they cannot distinguish the junior analyst from the senior partner doing related work, so any scenario allocated by exposure alone distributes displacement across the age structure of exposed occupations, which in the UK peaks among prime-age workers. Yet the emerging evidence points the other way: junior headcount contracts sharply at AI-adopting firms while senior staffing is unaffected, in UK firm data (Klein Teeselink, 2025), US resume histories (Hosseini and Lichtinger, 2026), and US payroll records (Brynjolfsson et al., 2025). Seniority bias, in other words, must be imposed on an exposure framework because it cannot be derived from one. We therefore add an age dimension to the displacement mechanics and use those three estimates as calibration anchors, asking how a youth-tilted shock changes the fiscal and distributional picture relative to age-neutral displacement—a correction that matters because younger workers sit differently in the household income distribution, interact differently with the benefit system, and face a longer horizon over which scarring can compound (Williamson et al., 2025).

The remainder of the report proceeds as follows. Section 2 reviews the literature on AI exposure measurement and the emerging evidence on labour-market effects. Section 3 describes the exposure mapping to SOC2020 major groups, the scenario design—the central calibration, the full scenario grid and the age-differentiated scenarios—and the PolicyEngine UK simulation framework. Section 4 presents results: the incidence of the shocks across the income distribution, the scenario grid, robustness checks, and the age dimension. Section 5 concludes with policy implications for retraining and lifelong learning and for the resilience of the tax base as income shifts from labour to capital, and Section 6 sets out limitations and further work.

2 Literature review

A large body of work on earlier general-purpose technologies documents that technological change over recent decades has been skill-biased and, more precisely, routine-biased: computerisation substituted for workers performing codifiable, routine tasks in the middle of the wage distribution while complementing non-routine cognitive work at the top (Autor et al., 2003). The distributional consequence was a widening of wage inequality and a polarisation of employment, documented early for Britain by Goos and Manning (2007) and shown to hold across sixteen European countries by Goos et al. (2014), while Acemoglu and Restrepo (2022) attribute much of the rise in US wage inequality since 1980 to task displacement. A notable contrast follows: where those earlier waves hollowed out the middle of the distribution, the AI exposure measures discussed below rise with income. Crucially, however, studies that trace these labour market shocks through to household disposable income find that tax-benefit systems absorbed much of the shock: progressive taxation and means-tested transfers substantially muted the pass-through of rising wage dispersion into household income inequality (Doorley et al., 2023). This distinction between market-income and disposable-income effects motivates the microsimulation approach adopted in this paper.

Artificial intelligence appears to break with the routine-biased pattern in an important respect. Occupation-level exposure scoring was pioneered by Frey and Osborne (2017), whose estimate that 47 per cent of US employment was at high risk of computerisation subsequent indices have refined. Measures constructed from the capabilities of modern AI systems—whether based on linkages between AI applications and occupational abilities (Felten et al., 2021), overlap between patent text and job task descriptions (Webb, 2019), or assessments of which tasks large

language models can perform (Eloundou et al., 2024)—consistently show that exposure rises with education and earnings. Professional, managerial and technical occupations score among the most exposed, a finding replicated for European labour markets (Williamson et al., 2025). Yet high exposure need not imply displacement: the same workers who are most exposed may also enjoy the largest complementarity gains, since AI can augment rather than replace complex cognitive work. Cazzaniga et al. (2024) formalise this ambiguity, showing that the net effect of AI on inequality depends on the balance between displacement and complementarity within highly exposed, high-income occupations, and can plausibly go in either direction.

Direct empirical evidence on the labour market effects of AI adoption remains limited and, where available, mixed. Firm-level studies generally find that AI adopters are larger, more productive and faster-growing than non-adopters, and that they pay higher wages, though it is difficult to separate the causal effect of adoption from selection into adoption by already-successful firms. Worker-level evidence is similarly inconclusive: some studies find employment growth at adopting firms alongside compositional shifts towards more skilled workers, while others detect reductions in hiring for exposed roles: US online-vacancy data show AI-exposed establishments posting AI jobs while curtailing non-AI hiring, with no detectable aggregate employment effect (Acemoglu et al., 2022), and Danish population-scale evidence finds only minimal effects of AI chatbots on earnings and hours so far (Humlum and Vestergaard, 2025)—a null result that usefully disciplines the magnitudes assumed in simulation exercises. This thin and heterogeneous evidence base is one reason scenario-based simulation, of the kind pursued here and in Doorley et al. (2026), remains a useful complement to ex-post estimation.

A related strand quantifies AI’s productivity potential at the task level. Eloundou et al. (2024) estimate that large language models, with appropriate complementary software, could enable roughly 15 per cent of all US worker tasks to be completed significantly faster at the same quality, with the share rising markedly once tooling built on top of the models is taken into account. Experimental and quasi-experimental studies of specific occupations point in a consistent direction: in programming, professional writing and customer support, access to generative AI tools raises output per worker—by around 14–15 per cent for customer-support agents (Brynjolfsson et al., 2025)—with the largest gains typically accruing to less experienced or lower-skill workers, thereby compressing within-occupation productivity gaps (Noy and Zhang, 2023). Whether this within-job compression translates into narrower wage dispersion depends on how the gains are shared and on which jobs survive.

Evidence specific to the United Kingdom is beginning to accumulate. The UK government’s official assessment of occupational exposure (DSIT and DfE, 2023), on which our exposure cross-walk draws, likewise places professional and managerial occupations among the most affected. Klein Teeselink (2025) link measures of generative AI exposure to UK firm-level employment records and find that more exposed firms reduced employment by around 4.5 per cent following the diffusion of generative AI, with losses concentrated in junior positions, which fell by roughly 5.8 per cent. From a macro-distributional perspective, Rockall et al. (2025) apply the IMF’s exposure-and-complementarity framework to the UK and conclude that, in contrast to earlier automation waves, AI could reduce wage inequality if displacement effects fall disproportionately on high-income workers—though sufficiently strong complementarity among those same workers could offset or reverse this compression. Our paper speaks directly to this open question by tracing both scenarios through the UK tax-benefit system.

The concentration of adjustment among junior workers found for the UK echoes emerging evidence of seniority-biased effects elsewhere. Using US resume and job-posting data, Hosseini

and Lichtinger (2026) document that employment of junior workers falls by around 9 per cent within six quarters of firms’ AI adoption, while senior employment is essentially unaffected, consistent with AI substituting for entry-level task content while complementing experienced judgement. In a similar vein, Brynjolfsson et al. (2025) find a 13 per cent relative decline in employment among workers aged 22–25 in the most AI-exposed occupations since late 2022, with older workers in the same occupations largely spared. If AI’s incidence is graded by seniority as much as by skill, the distributional consequences differ from those of a classic skill-biased shock, since junior workers are spread across the earnings distribution.

Finally, AI may reshape the distribution of income between factors as well as within labour. Task-based models of automation imply that the substitution of capital for labour in a widening range of tasks lowers the labour share and channels a growing portion of national income to the owners of capital (Acemoglu and Restrepo, 2018), and Moll et al. (2022) show theoretically that automation raises the returns to wealth itself. Because capital income is far more concentrated than labour income, this channel raises household inequality even if wage dispersion is unchanged; Korinek and Stiglitz (2019) develop the theory of such worker-replacing progress and the redistribution it calls for, and it strengthens the case examined by Bastani and Waldenström (2024) for progressive taxation of capital income as a policy response. Our simulations therefore consider not only wage scenarios but also the sensitivity of UK disposable-income inequality to shifts in the capital share.

3 Methodology

Our analysis transplants the methodology of Doorley et al. (2026) from Ireland to the United Kingdom. The approach proceeds in three stages: we first attach an occupation-level measure of exposure to artificial intelligence to each worker in a representative household survey; we then translate assumed economy-wide AI adoption effects into person-level shocks to employment, wages and capital income; and finally we pass both the baseline and the shocked data through a tax–benefit microsimulation model to recover the distributional and fiscal consequences. Each stage is described in turn, noting where UK data constraints force departures from the Irish implementation.

3.1 Measuring AI exposure

The starting point is the AI Occupational Exposure (AIOE) index of Felten et al. (2021), which links progress in ten AI application areas to the occupational ability requirements catalogued in O*NET, yielding a standardised score for each occupation that captures how much of its ability bundle overlaps with what AI systems can now do. Exposure alone, however, does not distinguish between occupations in which AI substitutes for workers and those in which it augments them. Following Doorley et al. (2026), we therefore adopt the complementarity adjustment of Pizzinelli et al. (2023), who construct an index $\theta_l \in [0, 1]$ from O*NET work-context descriptors (such as responsibility for others, physical proximity requirements, and the consequence of error) and job-zone preparation requirements. High- θ occupations are those in which AI is likely to complement human judgement rather than displace it. The complementarity-adjusted exposure measure (C-AIOE) shields high-complementarity occupations from the raw exposure score:

$$\text{C-AIOE}_l = \text{AIOE}_l \times (1 - (\theta_l - \theta_{\min})), \quad (1)$$

where $\theta_{\min}$ is the minimum complementarity across occupational groups, so that the least complementary group retains its full exposure score. Related occupation-level exposure measures include Tolan et al. (2021), the task-level GPT rubric of Eloundou et al. (2024), and the cross-country application in Cazzaniga et al. (2024); we retain the Felten–Pizzinelli combination for comparability with Doorley et al. (2026).

For the UK we build a bespoke crosswalk. Occupation-level AIOE scores mapped to the UK Standard Occupational Classification, following the DSIT/Felten mapping to SOC2020, are combined with complementarity indices θ that we reconstruct from the public O*NET 27.3 release using the recipe documented by Pizzinelli et al. (2023). Both series are then aggregated to the nine SOC2020 major groups (the one-digit level) using employment weights from ASHE 2025 Table 14. As in Doorley et al. (2026), the resulting C-AIOE scores are shifted so that the least-exposed major group scores exactly zero; under the allocation rule in Section 3.3 this group consequently receives no job losses, and the shift also prevents the negative values of the standardised index from corrupting the allocation weights.

One departure from the Irish study should be stated plainly. Doorley et al. (2026) operate at the two-digit ISCO level, giving roughly forty occupational groups over which exposure varies. The Family Resources Survey (FRS) person records used here carry only the one-digit SOC2020 code, so version 0.1 of this study distinguishes just nine major groups. All within-major-group variation in exposure is therefore lost, and the exposure gradient we impose on the population is correspondingly coarser than in the Irish original. Imputation of finer occupation codes from the Labour Force Survey is a planned refinement.

3.2 AI adoption scenarios

The size of the aggregate shock cannot be estimated from historical data and must be assumed. We follow the scenario architecture of Doorley et al. (2026). The central scenario adopts the widely cited estimates of Briggs and Kodnani (2023): generative AI displaces 7 per cent of employment, while the wages of workers who remain employed rise by 2.6 per cent in aggregate — the latter being the median wage-change estimate adopted by Doorley et al. (2026) from the literature they survey. Alongside the labour-market effects, AI adoption is assumed to raise the return to capital by 0.4 percentage points, lifting the baseline return of 1.005 per cent to 1.405 per cent.

Because the displacement and wage figures are genuinely uncertain, we complement the central scenario with a robustness grid spanning displacement rates from 0 to 10 per cent and aggregate wage growth from 0 to 5 per cent, a total of 66 scenario combinations; the capital-return shock is applied throughout the grid. The grid is anchored at its low end by Acemoglu (2025), whose task-based calculation implies modest aggregate employment effects, and at its high end by Brynjolfsson et al. (2025), whose evidence of substantial employment declines is specific to early-career workers in exposed occupations and should therefore be read as an upper bound rather than an economy-wide estimate.

3.3 Implementing the shock at the micro level

Employment shock. The aggregate number of displaced workers is the scenario displacement rate multiplied by (weighted) employee employment. This total is allocated across occupational

groups l in proportion to each group’s employment multiplied by its exposure:

$$\text{JobLoss}_l = \text{TotalJobLoss} \times \frac{\text{EMP}_l \times \text{C-AIOE}_l}{\sum_l \text{EMP}_l \times \text{C-AIOE}_l}, \quad (2)$$

so that job losses are concentrated in large, highly exposed groups, and the zero-scored least-exposed group loses no jobs. Within each group, the individuals who lose their jobs are selected by weighted random draws without replacement, with selection probabilities proportional to each individual’s exposure score times their survey weight, drawing until the group’s weighted quota in equation (2) is filled. Because the identity of the displaced is stochastic, all decile-level results are averaged over 50 seeded draws.

Displaced workers have their employment income set to zero and their labour-market status set to unemployed before the shocked simulation is run. Doorley et al. (2026) model the displaced as long-term unemployed — out of work for nine months or more, with entitlement to contributory unemployment benefits exhausted — so that they interact with the means-tested rather than the insurance tier of the benefit system. This contract is only partially expressible in PolicyEngine UK, which takes no unemployment-duration input; the shocked workers are treated as current-period unemployed and assessed by the model’s standard benefit rules. We document this as a limitation: to the extent that new-style (contributory) Jobseeker’s Allowance would in fact be exhausted after nine months, our results may modestly overstate the benefit support reaching displaced workers.

Wage shock. Workers who survive the employment shock share an aggregate wage-uplift pool equal to the scenario wage-growth rate times the surviving wage bill. The pool is distributed in proportion to complementarity, so that the wage change for group l satisfies

$$\text{WageChange}_l = \text{WageGrowth} \times \frac{\theta_l}{\bar{\theta}} \times \text{Wage}_l, \quad \bar{\theta} = \frac{\sum_l \text{EMP}_l \theta_l}{\sum_l \text{EMP}_l}, \quad (3)$$

where the normalisation by the employment-weighted mean complementarity $\bar{\theta}$ ensures the aggregate uplift equals the assumed wage growth. The economic logic mirrors equation (1): AI complements, rather than substitutes for, high- θ occupations, so those occupations capture a larger share of the productivity gains.

Capital shock. The 0.4 percentage-point increase in the return to capital is implemented by scaling each person’s interest and dividend income by the ratio of the shocked to the baseline return, $1.405/1.005 \approx 1.398$. Following Doorley et al. (2026), rental income is excluded from the scaling, on the grounds that the AI productivity channel operates through financial rather than housing capital. The self-employed are outside all three shocks, again as in the Irish study.

3.4 Microsimulation with PolicyEngine UK

Where Doorley et al. (2026) use the ESRI’s SWITCH model on SILC microdata, we use PolicyEngine UK (version 2.89.2), an open-source tax-benefit microsimulation model, running on person-level microdata from the Family Resources Survey 2024–25 obtained from the UK Data Service. Occupation codes are not carried in the processed PolicyEngine dataset, so we join the SOC2020 major group from the raw FRS `adult.tab` file onto the simulation’s person table using the identifier convention `person_id = SERNUM × 1000 + PERSON`, which matches every FRS adult. The simulation year is 2026.

The counterfactual design is a paired comparison on identical data: a baseline simulation and a shocked simulation are run on the same dataset, with the shocked run receiving the modified employment income, savings-interest income, dividend income and employment status of Section 3.3 as direct inputs. All reported effects are differences between the two runs, so sampling variation in the survey cancels and only the shock itself drives the results. We report four families of outcomes: the change in the government balance (the Exchequer cost of the shock, net of the additional revenue raised on higher wages and capital income); poverty rates before and after housing costs (BHC and AHC), person-weighted; the Gini coefficient of equivalised household net income; and mean changes in household net income by income decile, where deciles are fixed at their baseline positions so that individuals are tracked rather than re-ranked. On the choice of microsimulation platform and the UK tax–benefit modelling environment more generally, see Williamson et al. (2025).

4 Results

4.1 Employment, wage and capital income changes

We begin with the first-round labour market effects of the central scenario, in which 7 per cent of exposed employment is displaced and surviving workers in exposed occupations receive a 2.6 per cent wage uplift. Figure 1 plots the share of workers in each equivalised disposable income decile who transition from employment to unemployment. The gradient is strongly increasing: only 0.4 per cent of people in the bottom decile transition, rising steadily through the middle deciles to 5.0 per cent in the top decile. This reflects the occupational structure of AI exposure in the UK: the professional, associate-professional and administrative occupations most exposed to generative AI are concentrated in the upper half of the income distribution, whereas manual and personal-service occupations at the bottom are comparatively insulated. Displacement risk in this exercise is therefore a phenomenon of the top of the earnings ladder, not the bottom.

The wage side of the shock is, by contrast, almost distributionally neutral. Figure 2 shows the percentage change in employment income among workers who remain employed. Gains range from 2.45 per cent in the bottom decile to 2.66 per cent in the top—a spread of barely 0.2 percentage points around the 2.6 per cent uplift applied to exposed occupations. Because exposure is widespread across the deciles in which people work at all, the productivity dividend is spread thinly and evenly; it is the displacement margin, not the wage margin, that drives the distributional results below.

The third channel is capital income. Following the approach of Doorley et al. (2026), we assume the AI productivity gain is partly capitalised into returns on existing capital income, applied as a uniform proportional uplift (just under 40 per cent of the modelled productivity effect on capital returns). Although the percentage change is identical across deciles by construction, Figure 3 shows that the incidence in pounds is heavily concentrated: the top decile holds around 35 per cent of all capital income (some £8.0 billion of the aggregate in our data), the top two deciles together more than half, while the bottom decile accounts for under 3 per cent. Any capitalisation of AI gains therefore flows overwhelmingly to households that are already at the top of the distribution.

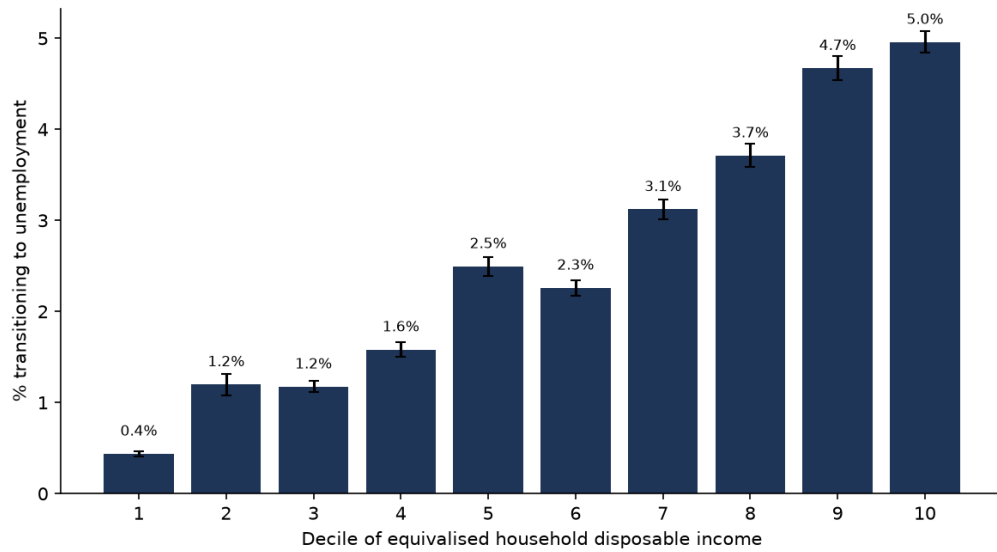


Figure 1: Share of the population transitioning from employment to unemployment by equivalised disposable income decile, central scenario. Error bars show 95 per cent confidence intervals across 50 seeded displacement draws.

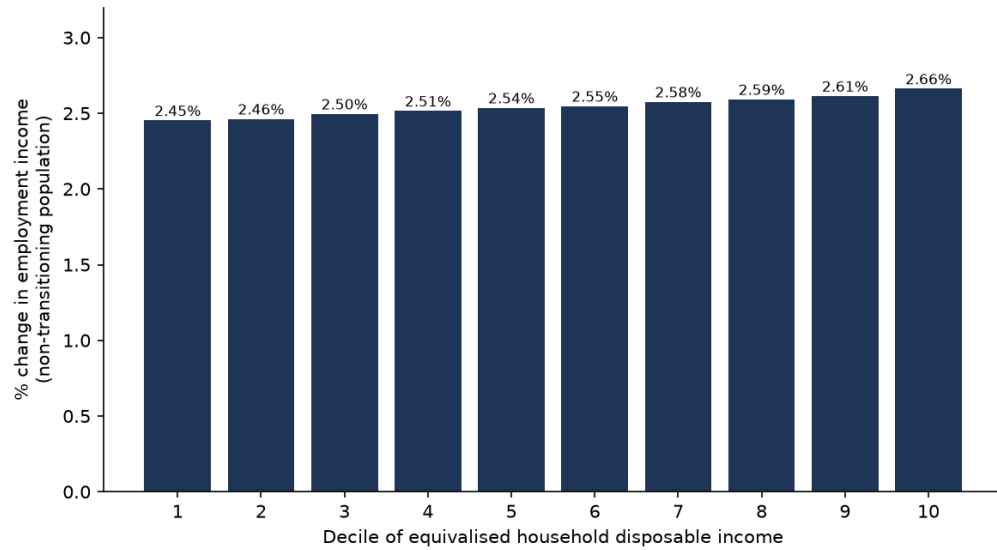


Figure 2: Percentage change in employment income among workers remaining in employment, by income decile, central scenario.

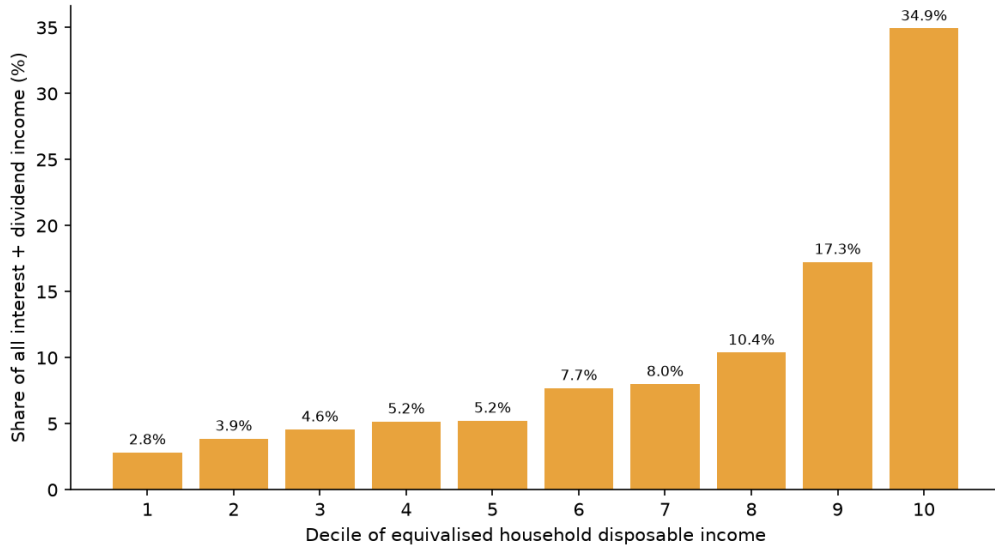


Figure 3: Capital income channel by decile: uniform proportional uplift, but pound gains concentrated where capital income is held.

4.2 Income distribution

Figure 4 decomposes the change in household income by decile into market income, benefits, and taxes and contributions, expressed relative to baseline disposable income. Market income losses are progressive in the literal sense: they are near zero in the bottom two deciles and grow to around 2 per cent of income in deciles 7–10, mirroring the transition gradient in Figure 1. The tax–benefit system then reshapes this pattern in two ways. Across the middle deciles, automatic benefit entitlements—Universal Credit in particular—rise by up to 0.7 per cent of income as displaced earners qualify for means-tested support, cushioning a substantial part of the market income loss. At the top, the cushioning comes instead through the tax system: households in deciles 7–10 see their tax and National Insurance liabilities fall by 0.2–0.6 per cent of income as high marginal-rate earnings disappear. Disposable income losses at the bottom of the distribution are well under 1 per cent of income, against 3–5 per cent in deciles 7–10.

The Irish results of Doorley et al. (2026) provide an external check on these numbers: with different microdata, a different tax-benefit model and a finer occupational classification, their central-scenario transition gradient runs from 0.7 per cent in the bottom decile to 5.3 per cent in the top, against 0.4 to 5.0 per cent here. That two independent implementations recover the same top-concentrated incidence suggests the gradient is a general feature of services-intensive economies facing generative AI, not an artefact of either study’s data or model.

4.3 Further scenarios

To move beyond the four headline presets we simulate a full grid of 66 scenarios, crossing eleven displacement rates (0 to 10 per cent of exposed employment) with six wage uplifts (0 to 5 per cent), each combined with the capital income channel. Figure 5 maps the resulting change in average household disposable income. The corners bracket the range: with a 5 per cent wage gain and no job loss, disposable income rises by 2.9 per cent; with 10 per cent displacement and no

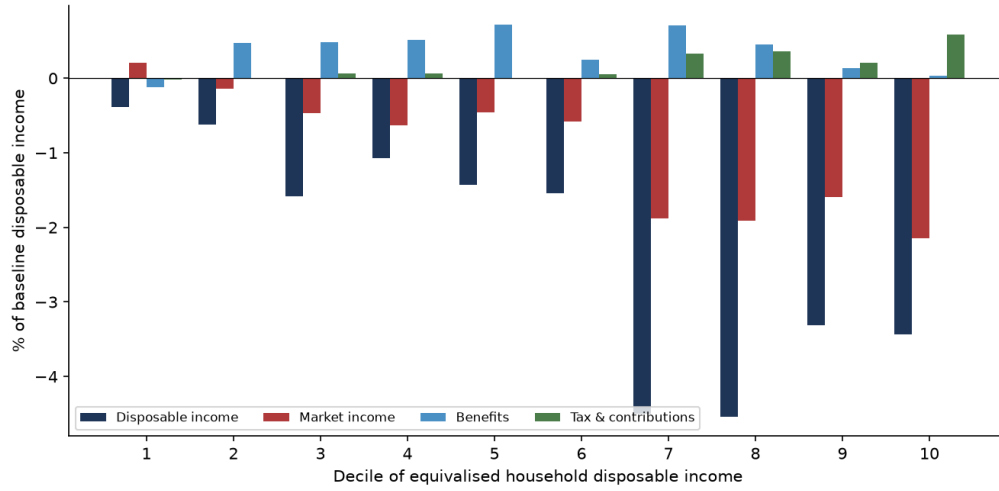


Figure 4: Decomposition of the change in household income by decile into market income, benefits, and taxes and contributions, central scenario, as a share of baseline disposable income.

wage gain, it falls by 5.5 per cent. The near-zero contour runs close to the grid diagonal—roughly one percentage point of wage growth offsets one percentage point of displacement in aggregate disposable income terms—so whether AI raises or lowers average living standards depends almost entirely on the balance struck between these two forces, about which the empirical literature remains genuinely uncertain.

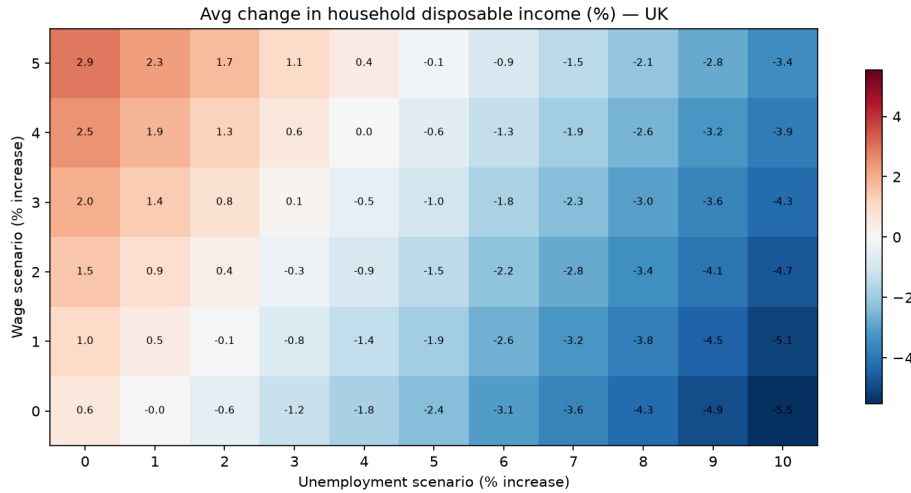


Figure 5: Change in average household disposable income (per cent) across the 66-cell grid of displacement rates and wage uplifts.

4.4 Exchequer cost

Figure 6 shows the corresponding fiscal picture. One measurement issue deserves note: in PolicyEngine’s UK aggregates, income tax plus National Insurance minus total benefit expenditure

(which includes the state pension) is a negative number, so expressing revenue changes relative to that net position produces uninformative percentages. We therefore express the net revenue change relative to gross income tax and National Insurance receipts. On that basis the grid spans a gain of 7.0 per cent of receipts (5 per cent wage growth, no displacement) to a loss of 10.2 per cent (10 per cent displacement, no wage growth). As with disposable income, the break-even locus lies close to the diagonal.

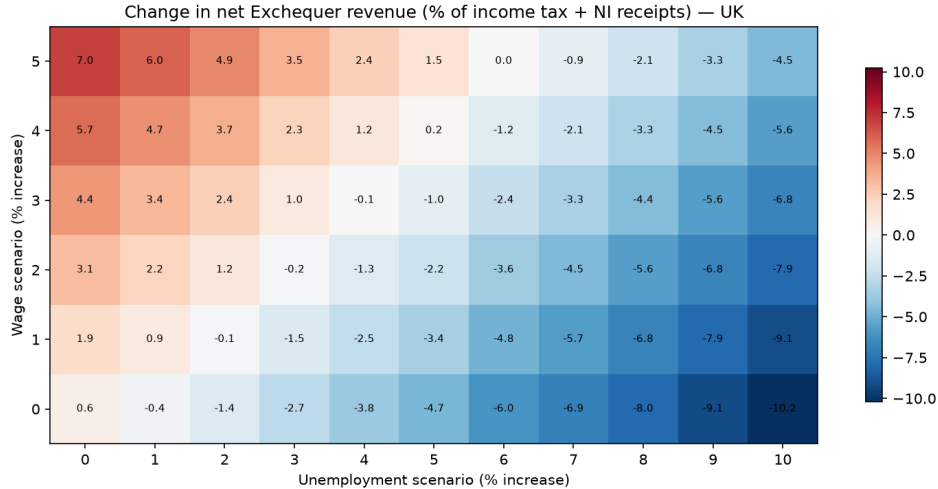


Figure 6: Net exchequer revenue change as a percentage of income tax and National Insurance receipts, across the 66-cell scenario grid.

Table 1 summarises the four preset scenarios. The annual exchequer cost ranges from £2.6 billion in the low scenario to £48.5 billion in the high scenario, with the central estimate at £18.6 billion—driven by lost income tax and National Insurance on displaced high earners together with higher benefit spending. Poverty (before housing costs) rises by 0.16 percentage points in the low scenario, 2.1 points in the central case and 4.1 points in the high case; the Gini coefficient rises by 0.2, 1.3 and 2.3 points respectively. Note that PolicyEngine’s before-housing-costs poverty line is an *absolute* threshold, so these poverty changes measure absolute impoverishment rather than movement relative to a shock-shifted median. Repeating the central scenario over twenty independent displacement draws gives a mean exchequer cost of £18.6 billion (standard deviation £1.7 billion, range £15.6–21.6 billion), a poverty increase of $+1.99 \pm 0.23$ percentage points, and a Gini increase of $+1.15 \pm 0.11$ points, so the qualitative conclusions are not artefacts of any single draw.

Table 1: Summary of preset scenarios (single draw, seed 0)

Scenario	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty (BHC) (pp)	Δ Gini (pp)
Low (1% displacement, no wage gain)	0.22	2.6	+0.16	+0.19
Central (7%, 2.6% wage gain)	1.61	18.6	+2.10	+1.32
High (13%, 2.6% wage gain)	2.98	48.5	+4.12	+2.29
Central, youth-tilted	1.60	17.5	+2.32	+1.36

4.5 Headline income inequality

Figure 7 maps the change in the Gini coefficient of equivalised disposable income across the grid. The striking result is its sign: the Gini rises in all 66 scenarios, from +0.05 percentage points in the capital-only cell (no displacement, no wage gain) to +1.9 points at 10 per cent displacement with a 5 per cent wage uplift. This exceeds the Irish range of roughly +0.3 to +1.2 points reported by Doorley et al. (2026), but the mechanism is the same and is one of polarisation rather than a simple rich-get-richer story: displacement strikes at the top of the earnings ladder and pushes affected households down the distribution, while the wage uplift lifts the surviving employed—who are disproportionately in the upper-middle and top deciles—further away from workless and low-income households. The capital income channel reinforces this at the very top. Because both the loss and the gain sides of the shock widen gaps between households, no combination of parameters within the grid reduces measured inequality.

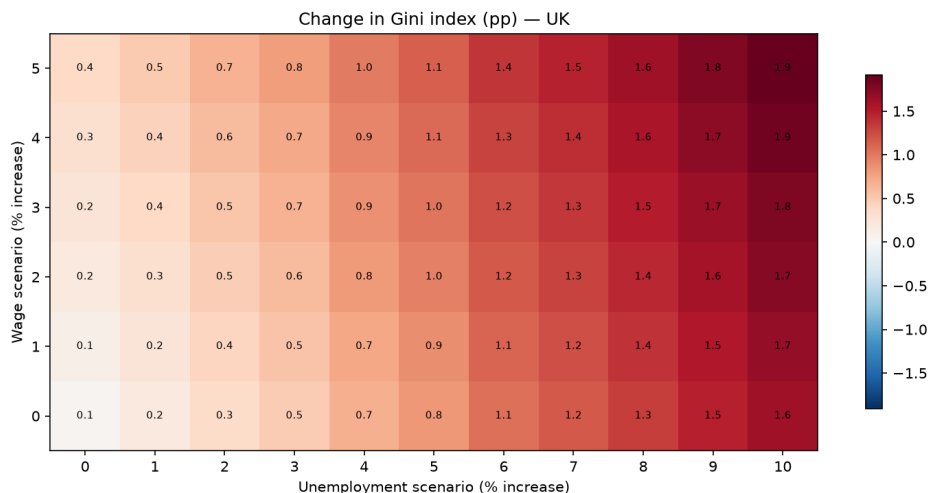


Figure 7: Change in the Gini coefficient of equivalised disposable income (percentage points) across the 66-cell scenario grid.

4.6 Robustness

Four checks probe the sensitivity of these findings. First, the choice of exposure index is immaterial: rerunning the central scenario with the DSIT AIOE mapping and with the Eloundou et al. (2024) GPT-exposure beta in place of the C-AIOE leaves the top-to-bottom transition gradient essentially unchanged (top-decile transition shares of 4.9 and 4.7 per cent against 4.9 under C-AIOE; bottom-decile shares of 0.3–0.4 per cent under all three) and moves the exchequer cost by less than £1.2 billion.

Second, a uniform comparator in the spirit of Doorley et al. (2026) isolates what the exposure gradient contributes: allocating the same aggregate displacement, wage and capital shocks uniformly across occupations yields a *lower* exchequer cost (£14.2 billion against £18.6 billion, because uniformly drawn displaced workers earn and therefore pay less tax) but a *larger* poverty increase (+2.31 against +2.10 points) and a larger Gini rise (+1.46 against +1.32 points). The concentration of AI exposure among high earners thus makes the shock fiscally more expensive but distributionally less regressive than an occupation-neutral shock of the same size.

Third, the occupational structure that drives everything is close to its administrative benchmark: the FRS employment shares across the nine SOC2020 major groups differ from the ASHE 2025 jobs distribution by at most 1.6 percentage points in any group.

Fourth, Monte Carlo variation across displacement draws (reported with Table 1) leaves all qualitative conclusions intact.

4.7 The age dimension and seniority-biased scenarios

We extend the Irish framework along a dimension it does not explore: the age incidence of displacement. Recent empirical work suggests that early-career workers bear a disproportionate share of AI-driven job loss (Brynjolfsson et al., 2025; Hosseini and Lichtinger, 2026). In our central scenario, which allocates displacement purely by occupational exposure, workers aged 16–24 account for only 3.0 per cent of the displaced. The youth-tilted preset reweights displacement probabilities towards junior workers while holding the aggregate displacement rate at 7 per cent; this raises the 16–24 share to 4.8 per cent. The exchequer cost falls slightly (£17.5 billion against £18.6 billion, since younger displaced workers pay less tax), but poverty rises by more: +2.32 percentage points against +2.10, because young workers have thinner savings and benefit entitlements and are more likely to sit in households near the poverty line.

Table 2 reports three scenarios anchored directly to headline estimates in the recent empirical literature, translated to the UK employment structure: the occupational displacement estimates of Klein Teeselink (2025), the entry-level hiring contraction of Hosseini and Lichtinger (2026), and the early-career employment declines in AI-exposed occupations documented by Brynjolfsson et al. (2025).

Table 2: Paper-anchored displacement scenarios, UK translation

Scenario	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty (BHC) (pp)
Klein Teeselink (2025): −4.5% economy-wide ^a	1.03	20.6	+1.75
Hosseini and Lichtinger (2026): −9% juniors, at-adopter ^a	0.10	1.0	+0.11
scaled by adopter employment share (16%)	0.02	0.1	+0.03
Brynjolfsson et al. (2025): −16% ages 22–25 ^{a,b}	0.02	0.1	+0.02

^aSource estimates are firm-conditional or relative declines; applying them economy-wide (or as absolute rates) makes these upper bounds. ^bA *relative* decline versus less-exposed groups in the November 2025 version of the paper (earlier drafts reported 13 per cent).

Two lessons emerge. First, cohort-specific shocks are fiscally small. The Hosseini and Lichtinger (2026) and Brynjolfsson et al. (2025) scenarios, which confine displacement to entry-level and early-career workers, displace 0.1 million and 0.02 million workers respectively and cost the exchequer £1.0 billion and £0.1 billion a year—an order of magnitude or two below the broad-based scenarios—precisely because junior workers contribute little income tax and National Insurance. Only the economy-wide Klein Teeselink (2025) scenario, with just over one million displaced, generates fiscal and poverty effects comparable to our central preset. Second, seniority bias must be imposed; it cannot be derived from occupational exposure alone. Even with the junior tilt, displacement rates by age remain highest for prime-age workers in the Klein Teeselink (2025) scenario—6.0 per cent of employed 35–44-year-olds are displaced, against 1.7 per cent of 16–24-year-olds—because young people in the UK are concentrated

in low-exposure service occupations. If AI adoption in practice falls hardest on juniors, as the hiring-margin evidence suggests, that pattern operates through recruitment channels that exposure-based simulation frameworks such as ours, and Doorley et al. (2026), do not capture by construction.

5 Discussion and policy implications

The results presented above should be read as scenario analysis rather than prediction. We do not attempt to forecast the pace or extent of AI adoption in the United Kingdom; instead, we trace the distributional and fiscal consequences that would follow *if* labour-market shocks of a given size and occupational incidence were to materialise. The value of the exercise lies in the comparative statics: how the UK tax-benefit system, as legislated, would mediate a range of plausible AI-driven wage and employment shocks, and where the pressure points in that system lie. The scenario parameters themselves are anchored in the empirical and modelling literature—including Acemoglu (2025), Briggs and Kodnani (2023) and Cazzaniga et al. (2024)—rather than estimated from UK data, and the wide span between our optimistic and pessimistic scenarios is a deliberate acknowledgement of the uncertainty surrounding all of these anchors.

A first, and reassuring, finding is that the UK tax-benefit system provides substantial automatic insurance to low-income households. Across all scenarios, means-tested support—principally Universal Credit—absorbs a large share of the market-income losses experienced by households in the lower deciles, while progressive income taxation claws back a portion of the losses higher up the distribution before they reach disposable income. This cushioning is not free. In our central scenario the combined cost to the exchequer of lower receipts and higher benefit expenditure amounts to £18.6 billion per year, and the fiscal cost scales with the severity of the shock. The distributional protection that households experience is, in effect, purchased through a deterioration of the public finances, and any assessment of the system’s resilience to AI-driven disruption must weigh the two together.

A second finding is less comfortable: disposable-income inequality rises in every scenario we examine, even though the incidence of the simulated losses is, in proportional terms, progressive—displacement risk rises monotonically with household income (Figure 1), and average losses are largest at the top. The mechanism is polarisation *within* the distribution rather than a transfer between its top and bottom. Each displaced worker falls from somewhere on the earnings ladder to the bottom of it, while the wage and capital gains lift those who remain; because the displaced come disproportionately from high-income households, the shock simultaneously thins the upper-middle of the distribution and swells its lower tail. When households are re-ranked on their post-shock income, as the Gini index does, dispersion increases even though the decile-by-decile average losses, computed at fixed baseline ranks, look progressive. This is the same re-ranking mechanism Doorley et al. (2026) identify for Ireland, and it differs from earlier automation waves, in which exposure fell on mid- and low-wage routine work (Autor et al., 2003; Doorley et al., 2023).

Three policy implications follow. First, because the modelled losses fall on identifiable occupational groups rather than diffusely across the workforce, retraining and lifelong-learning provision targeted at workers in highly exposed occupations is the most direct lever available; the tax-benefit system can replace income but cannot restore occupational attachment. Second, the emerging evidence that generative AI operates as a seniority-biased technology—compressing demand for junior and entry-level roles in exposed occupations (Hosseini and Lichtinger, 2026;

Brynjolfsson et al., 2025), with early UK-specific corroboration in Klein Teeselink (2025)—argues for close monitoring of youth labour-market entry. Our youth-tilted scenario is a first step in this direction, but it can only re-weight who is displaced within a single year: the dynamic costs of a scarred entry cohort — foregone experience, delayed progression, persistent earnings losses — lie outside a static framework and mean even the tilted scenario understates the long-run burden on new entrants. Third, the fiscal results expose a structural vulnerability: the UK tax base leans heavily on the taxation of labour income through income tax and National Insurance. If AI shifts the functional distribution of income from labour towards capital, the base that currently finances the cushioning we document would itself erode, a possibility that has motivated recent work on the taxation of capital in an AI-intensive economy (Bastani and Waldenström, 2024). Ensuring the resilience of revenues under such a shift is a first-order question that lies beyond the static incidence analysis presented here, but our results make clear why it matters.

Taken together, the results suggest that the United Kingdom enters a period of potential AI-driven labour-market disruption with a tax-benefit system that redistributes the losses effectively but does not, and cannot, prevent them. The system converts a regressive market-income shock into a smaller, more progressive disposable-income shock at a substantial and shock-dependent fiscal cost, while the polarising character of AI exposure pushes measured inequality upwards regardless. Policy attention is therefore best directed at the margins the tax-benefit system does not reach: the occupational reallocation of exposed workers, the entry prospects of young workers, and the long-run robustness of a labour-taxation-heavy revenue base.

6 Limitations and further work

Several limitations qualify the results, most of them stemming from data constraints rather than modelling choices, and several of them defining our agenda for further work.

First, occupational exposure is assigned at the 1-digit SOC level, a coarser join than the 2-digit ISCO classification used by Doorley et al. (2026) for Ireland. The Family Resources Survey records occupation only at the major-group level, so within-group heterogeneity in AI exposure—which is substantial in the exposure indices of Felten et al. (2021), Webb (2019) and Eloundou et al. (2024)—is averaged away. We plan to address this with a quantile-forest imputation of detailed occupation, trained on the 4-digit occupational codes available in the Labour Force Survey and conditioned on characteristics observed in both surveys, which would allow exposure to be assigned at a resolution comparable to, or finer than, the Irish study.

Second, we use the uncalibrated FRS survey weights rather than PolicyEngine’s enhanced dataset. The enhanced dataset improves aggregate fiscal accuracy through reweighting and household cloning, but the cloning procedure breaks the one-to-one correspondence between records and observed occupations on which our exposure join depends. Aggregate fiscal magnitudes, including the £18.6 billion central-scenario cost, should therefore be read with the usual caveats attaching to unadjusted survey grossing.

Third, the modelling of displacement is deliberately simple: displaced workers are treated as unemployed in the current period, with earnings set to zero and benefit entitlements recomputed under existing rules. The duration structure of unemployment is not modelled, so the exhaustion of contributory benefits—new-style Jobseeker’s Allowance is payable for at most six months—cannot be expressed within our static framework. Over horizons longer than the contributory window, our results will overstate the support available to displaced workers without means-tested entitlements.

Fourth, the self-employed fall outside all of our shock scenarios. Exposure indices and the displacement literature on which our parameters rest are defined over employees, and the earnings dynamics of self-employment under AI adoption—which may include both displacement and new opportunities—are sufficiently different that we exclude the group rather than impose an arbitrary treatment. Given that self-employment accounts for a meaningful share of UK employment, and is itself concentrated in some exposed occupations, this is a substantive omission.

Fifth, a computational caveat: results reported on the scenario grid rest on a single seeded random draw of which exposed workers are displaced, whereas the decile-level results average over 50 draws. The grid figures therefore carry Monte Carlo noise that the headline distributional results do not, although sensitivity checks suggest the aggregate magnitudes are stable across seeds. Finally, and most fundamentally, the scenario parameters are anchors taken from the international literature rather than elasticities estimated on UK data. As UK-specific evidence accumulates—early examples include Klein Teeselink (2025) and Rockall et al. (2025)—the scenarios can be re-anchored, and the gap between scenario analysis and conditional forecast will narrow. Extending the exposure measurement, restoring calibrated weights through an occupation-preserving reweighting, and introducing benefit-duration dynamics are the immediate priorities for further work.

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